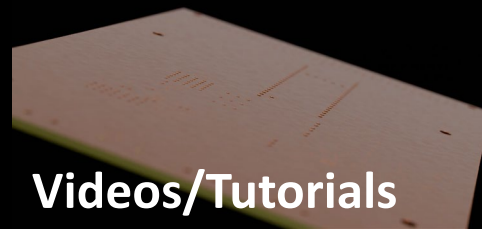
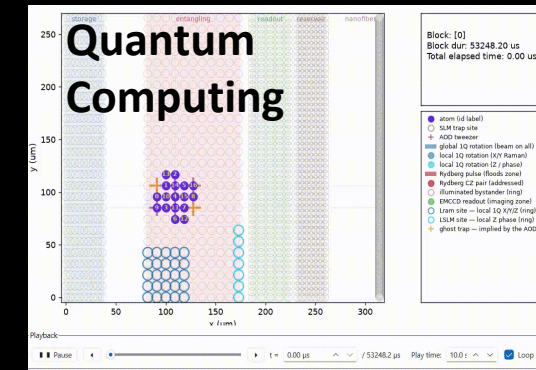
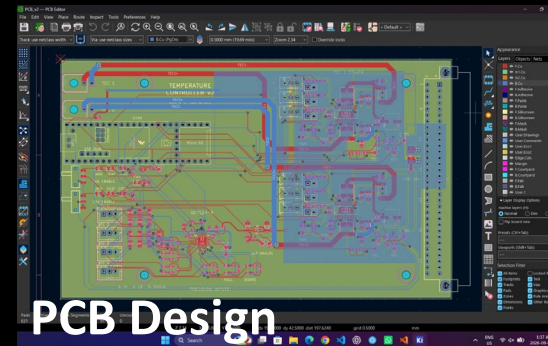


Modern Experimental Techniques

3) Getting things made

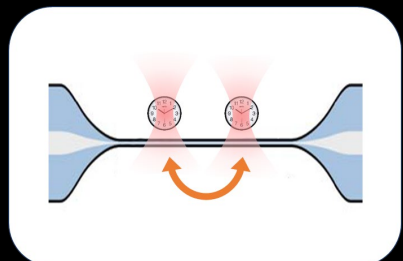


Videos/Tutorials



Shayne Bennetts

s.p.bennetts@g.iams.sinica.edu.tw



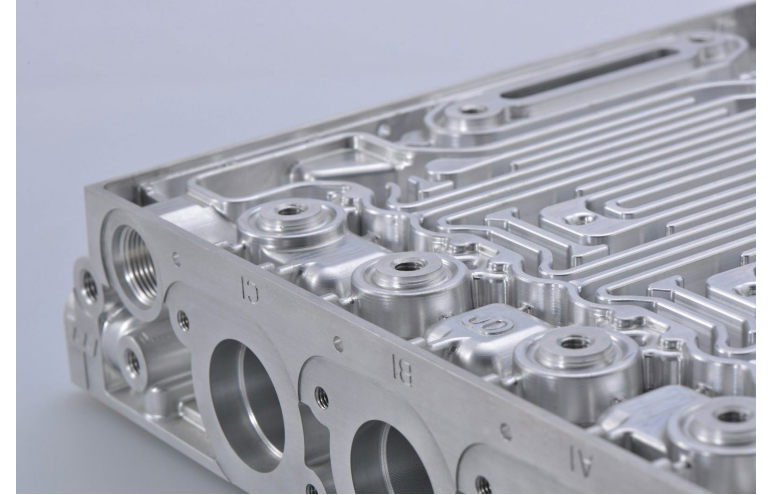
IAMS Yb Lab
Institute of Atomic and Molecular Sciences
Academia Sinica, Taiwan

<https://iamsquantum.github.io>



- Laptop with VS Code and Claude Code signed in
- OpenSCAD and Bambu Studio installed and opened once
- /tutor L3 in Claude Code: it updates the repository and installs what is missing
- A USB-C data cable for the ESP32: the instructor has spares
- WiFi details in the room
- Housing files due Fri 9 Oct, 2 pm



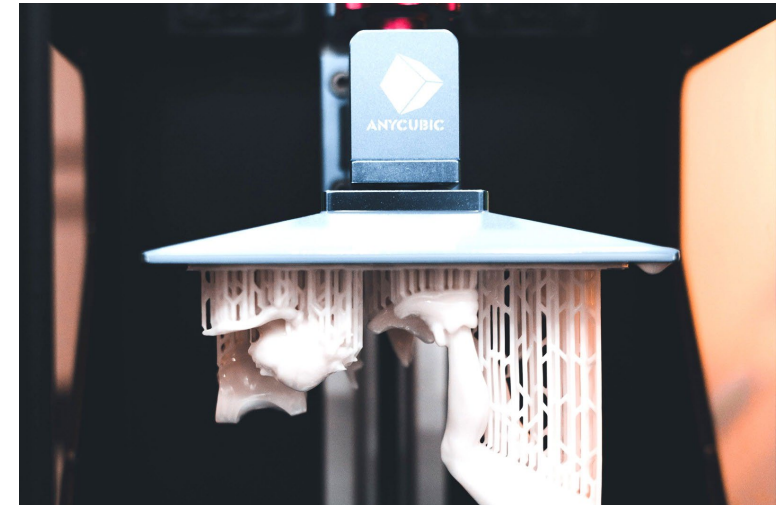
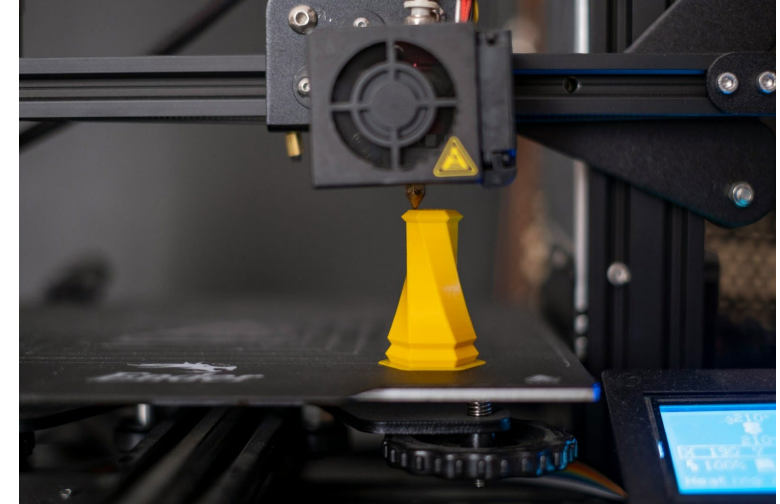


- Here, this week: the Bambu P1S, the laser cutter, hand tools
- Ordered: CNC machining, sheet metal, resin and metal printing, EDM
- A box: print it. A plate: cut it. Metal: order it.

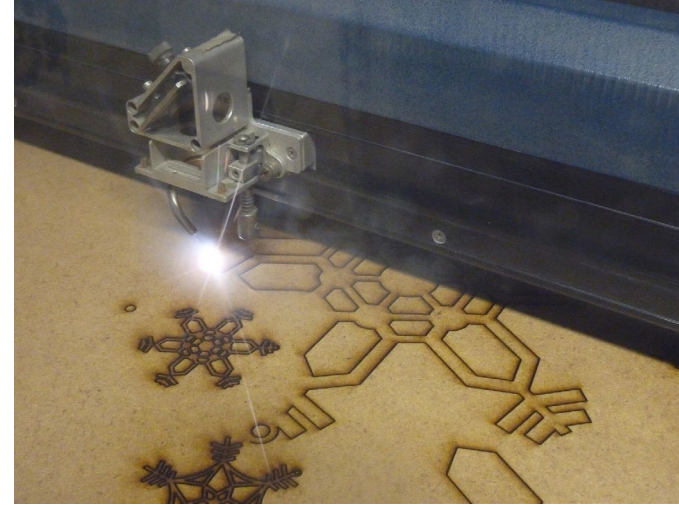


- A hand drill opens a printed hole out to size
- A tap cuts an M3 thread: 2.5 mm hole, turn, back off, turn
- Files take a burr or a tenth of a millimetre off an edge
- A hacksaw cuts rod, tube and acrylic sheet to length
- Everything bigger is ordered: a STEP file and a drawing

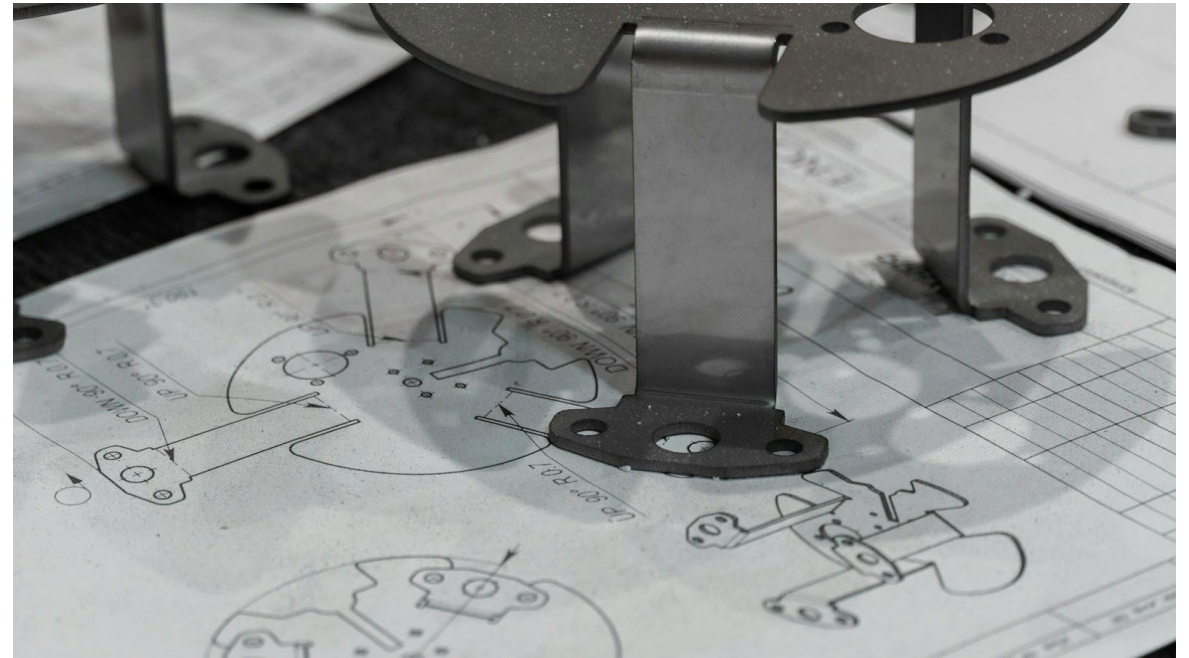
- FDM (Bambu P1S): plastic laid down in layers, a 256 mm cube
- 2 mm walls, no fine detail; a few dollars, printed overnight
- Resin (Phrozen): 25–50 μm layers, smooth, fine detail; brittle
- Resin parts are washed in alcohol and cured under UV; gloves on
- Metal printing exists too: steel, aluminium, titanium, from a service



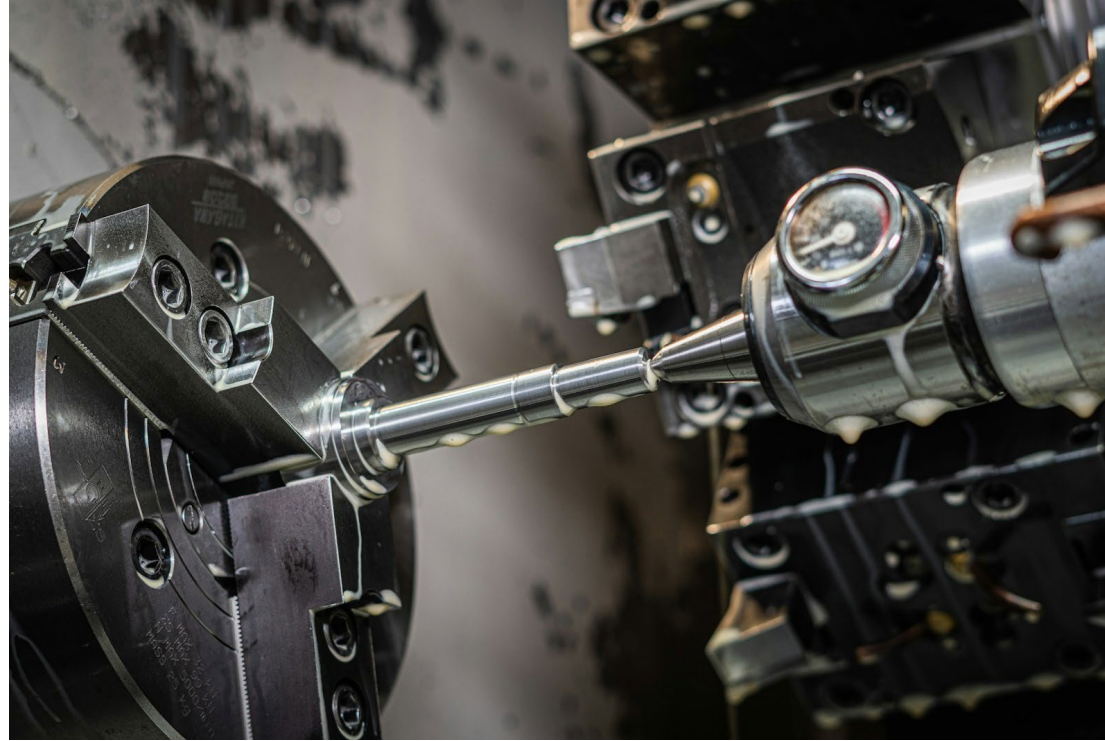
- Acrylic 2–10 mm and plywood from a DXF; the instructor runs our cutter
- Kerf about 0.2 mm: the beam burns away a thin line
- Panels, lids, windows, boxes of slotted plates
- Too thick or too hard for the laser: a water-jet, kerf about 1 mm; ordered
- Thin steel cheaply: order the pattern as a JLCPCB stencil



- Aluminium or steel 1–3 mm, cut flat, bent on a press brake
- Inside bend radius at least the sheet thickness
- Ordered from a service: cut and bent in days, a few dollars a part



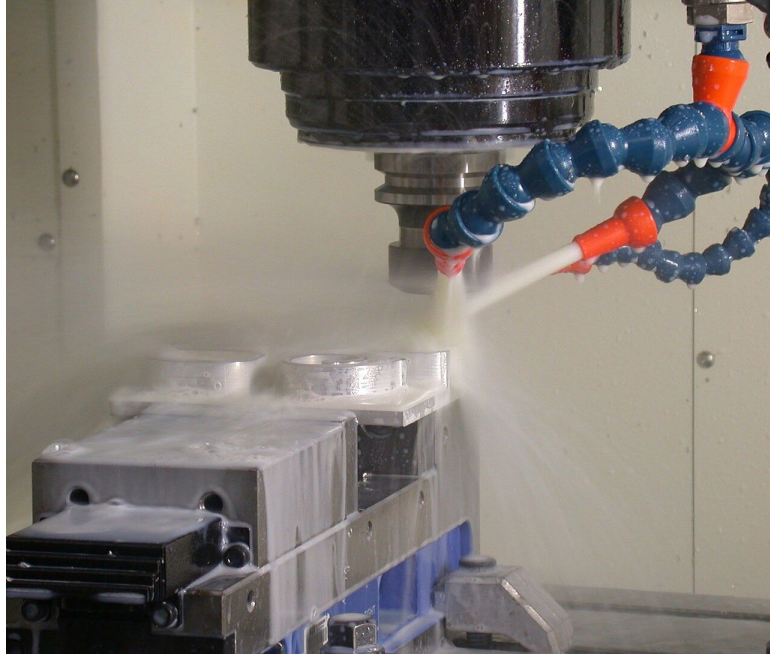
TiGP Turning: anything round



- Shafts, bushings, flanges, adapters, stand-offs
- Grooves, chamfers and threads come almost free
- One axis: the part spins, the tool moves

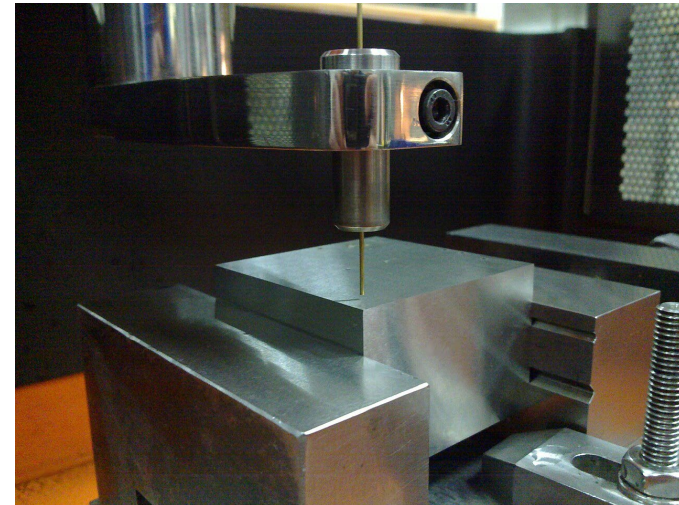
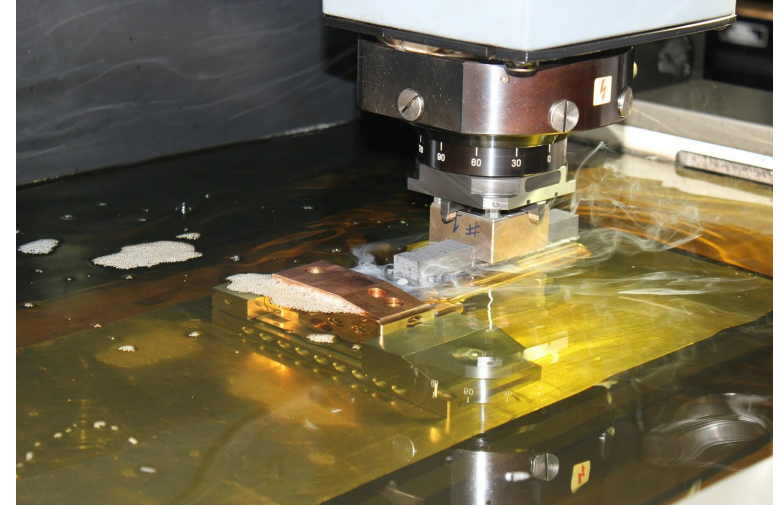


- About ± 0.1 mm as standard; tighter costs more
- Everything from one side, or a second set-up
- Inside corners have a radius: bigger is better



- Upload a STEP, pick the material, a price in minutes
- Typically 2–4 weeks and US\$100s; tolerance about 0.1 mm as standard
- Same rules as the manual mill: tool radius, reached from above
- 5-axis reaches around a part; EDM cuts sharp corners

- Wire EDM: a thin wire erodes any conductive metal, even hardened
- Inside corners down to the wire radius, about 0.1 mm
- EDM drilling: 0.3–3 mm holes through thick steel; a start hole
- Slow and precise, ± 0.005 mm; a service, with the same STEP or DXF



Process	Materials	Tolerance	Best for	Time and cost
FDM print (here)	PLA, PETG, ASA	about ± 0.2 mm	boxes, brackets, jigs	hours, a few US\$
Laser cut (here)	acrylic, plywood	about ± 0.2 mm	panels, lids, windows	minutes
Mill, lathe (ordered)	aluminium, brass, steel	about ± 0.1 mm	faces, pockets; round parts	2–4 weeks, US\$100s
Water-jet (ordered)	metal, glass, stone	about ± 0.2 mm	thick flat plates	weeks
Resin print (ordered)	photopolymer resin	about ± 0.1 mm	small, fine, smooth parts	days to weeks
Metal print (ordered)	steel, Al, titanium	± 0.2 mm, then machined	channels, lattices	weeks, US\$100s

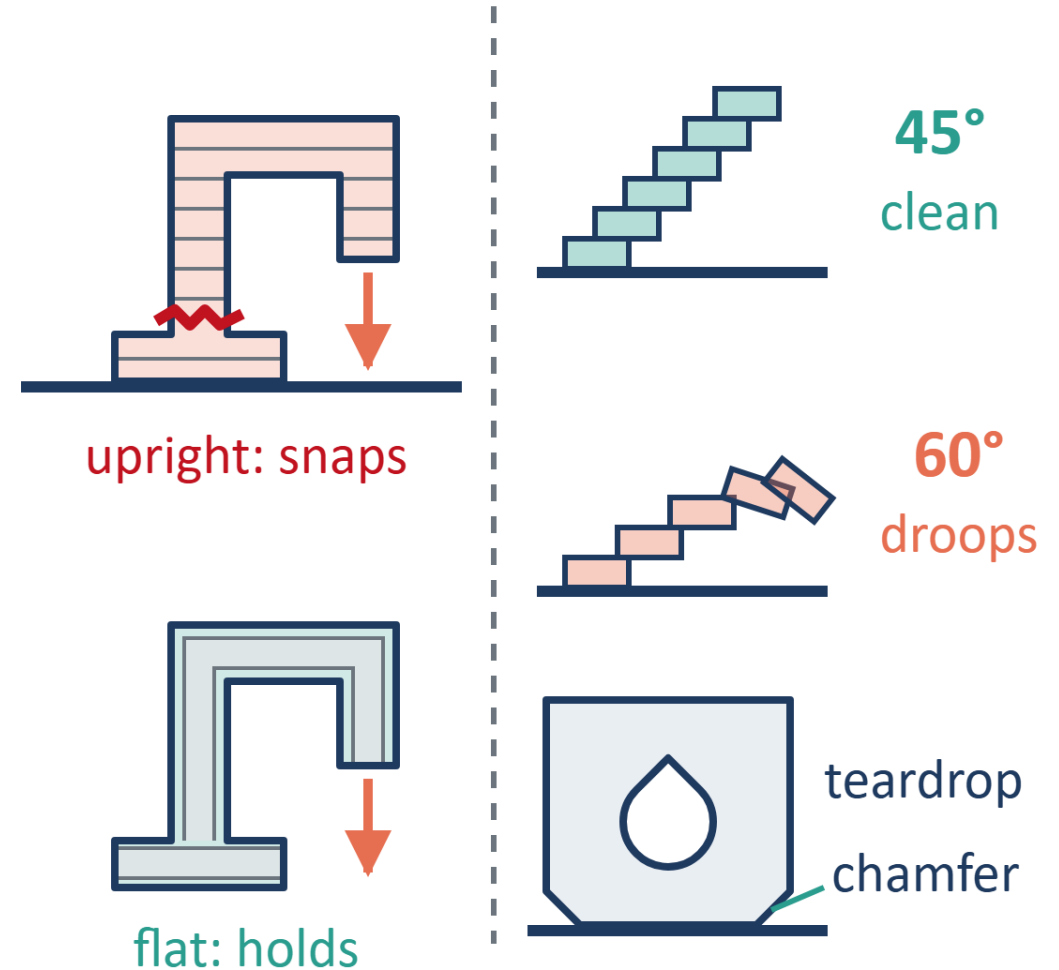
- Pick the process by the material and the tolerance you need
- Tolerances are typical values; the quote states yours

Tips to make 3D prints and cut parts turn out right

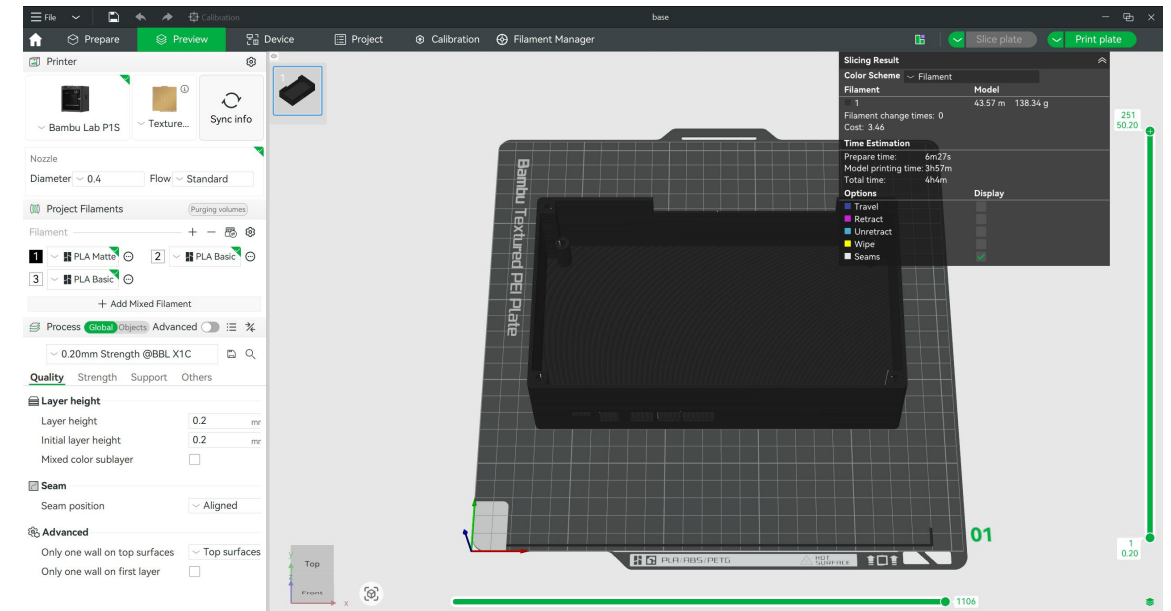
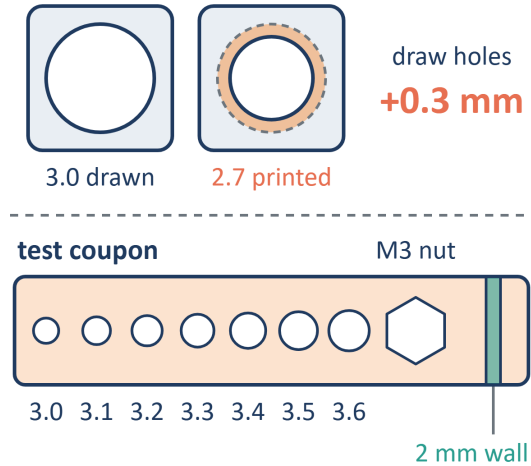
Orientation · overhangs · walls · fits ·
fasteners · materials · slicer



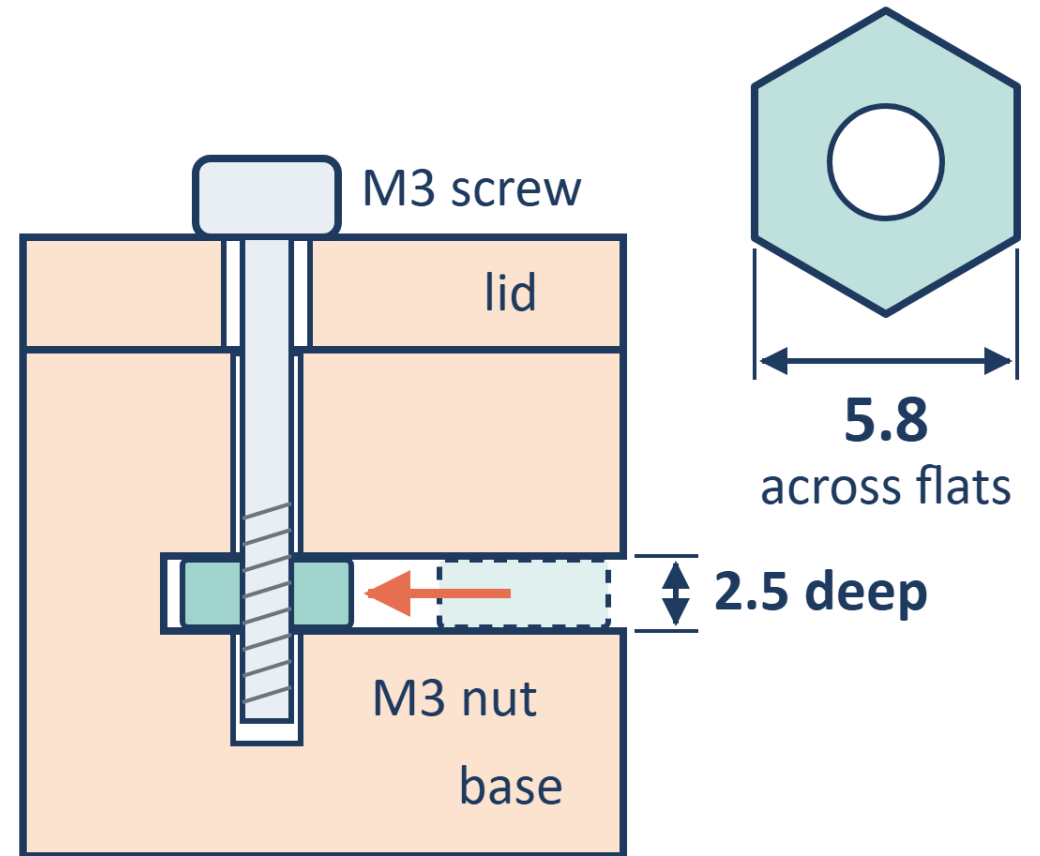
- Strong along the layers, weak across them (about half): print flat
- A clip or snap-fit must flex in the plane of the layers
- Up to about 45° from vertical prints on air; steeper droops
- Design out supports: re-orient, split, chamfer, teardrop holes
- Supports cost time, plastic and a rough surface



- Strength comes from the walls, not the infill: model 2 mm walls
- Infill 15–25 % gyroid; 100 % is wasted plastic; layers 0.2 mm
- A 0.5 mm chamfer at the bed hides the elephant's foot
- Holes print small, pegs large: draw holes +0.3 mm, slots +0.2
- Sliding fit 0.2–0.3 mm gap; press fit 0.05–0.1 mm
- Print a test coupon first: holes 3.0–3.6, a nut pocket, a wall



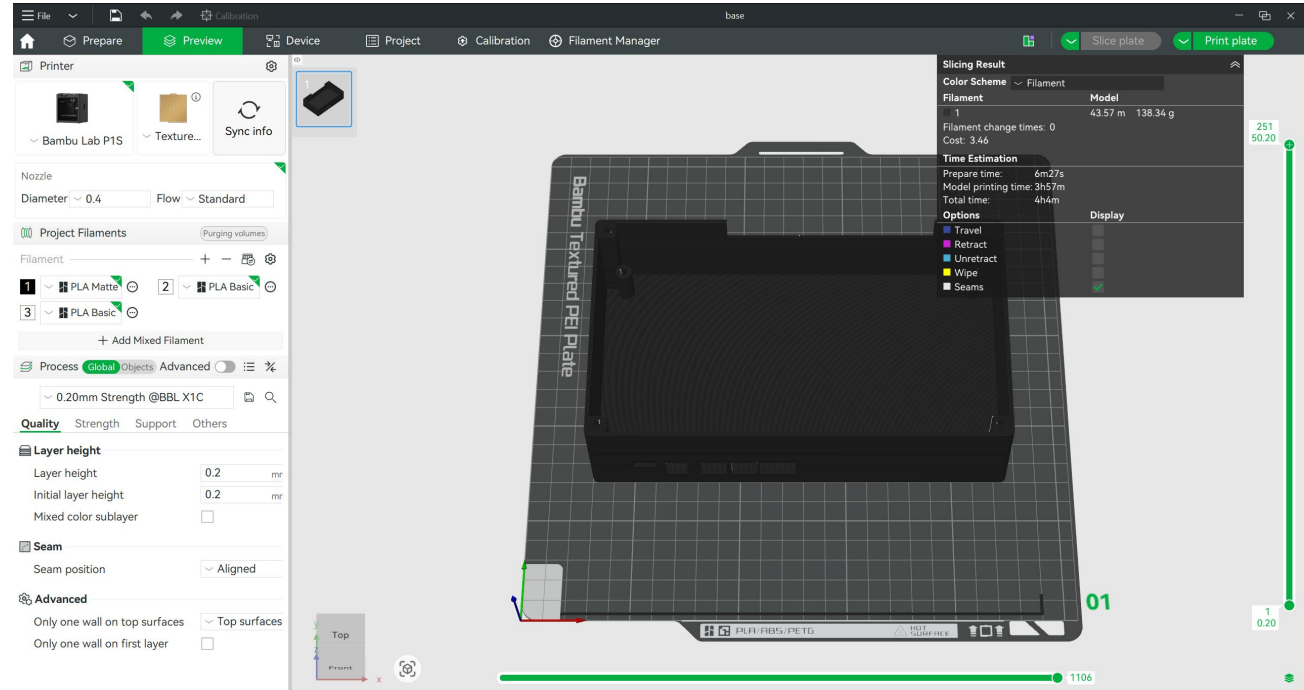
- Class standard: an M3 nut in a hexagonal pocket
- Pocket 5.7–5.8 mm across flats, 2.5 mm deep
- Clearance hole 3.4 mm; printed, draw it 3.7 mm
- The pocket opens to the side; the screw pulls the nut in
- Heat-set insert: brass, melted in with a soldering iron
- Metal parts: a tapped thread, named on the drawing
- Never print a thread below about M8



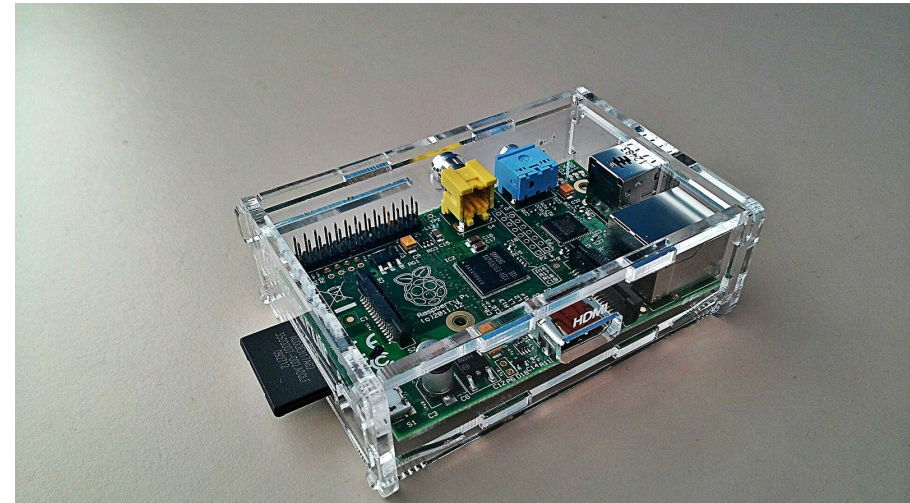
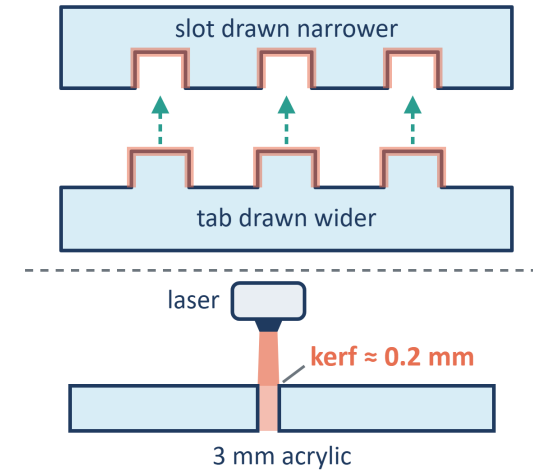
- PLA: stiff, easy, softens at about 60 °C
- PETG: tougher, a little flexible, about 80 °C
- ASA or ABS: about 100 °C, UV-stable; needs an enclosed printer
- Warping plastics want an enclosure and glue stick on the bed
- TPU rubber-like; PA-CF strong, hardened nozzle; PC about 120 °C
- Colour is free; multi-colour needs the AMS feeder



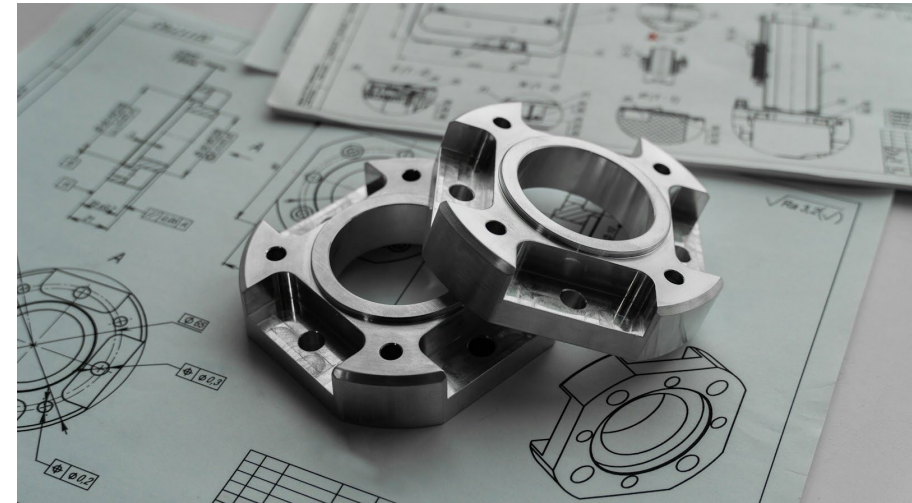
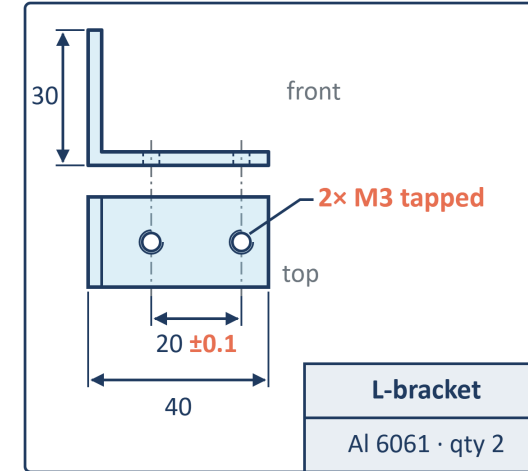
- Import the STEP or STL into Bambu Studio
- Lay the flat face down on the plate
- Look at the supports: many means turn the part or redesign
- Slice: read the print time and the grams
- Export the 3MF; the class queue is printed after the 9 Oct cutoff
- The plate is 256 × 256 mm: our 190 × 110 mm box fits



- Acrylic 3 mm for lids and panels
- One DXF per plate, in millimetres, closed outlines
- Kerf: draw slots narrower and tabs wider by about 0.2 mm
- Engrave labels in the same run
- Cast acrylic engraves white; extruded is cheaper
- A cut lid over a printed base is a good pairing



- Inside radii at least the tool; start from stock sizes
- Dimension from the face the machinist will measure from
- Tolerance only what must fit: ± 0.1 mm is standard, tighter costs more
- Name threads: M3 $\times$ 0.5 tapped, 6 mm deep
- Hand over a STEP, a drawing and what the part is for
- Ask before cutting: can you hold this? which side first?

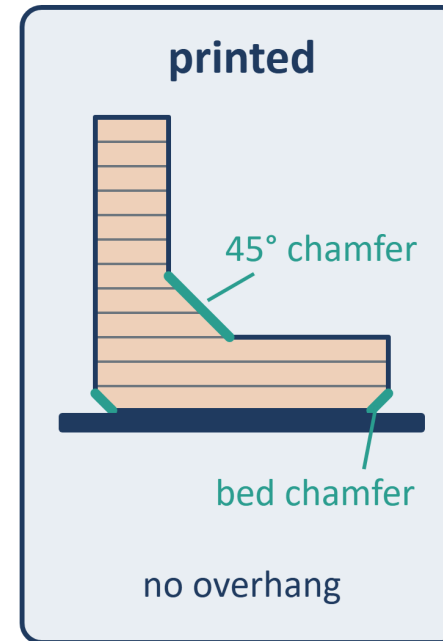
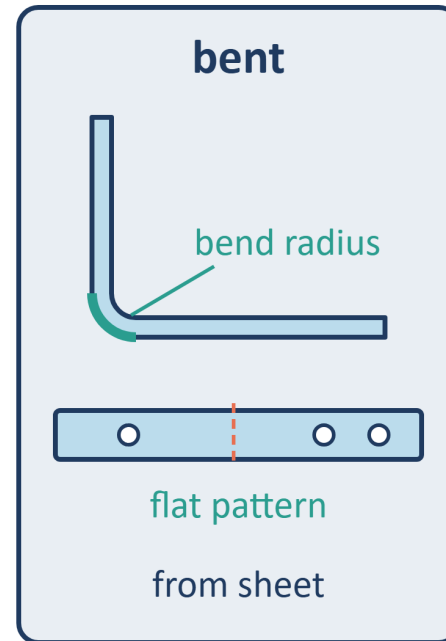
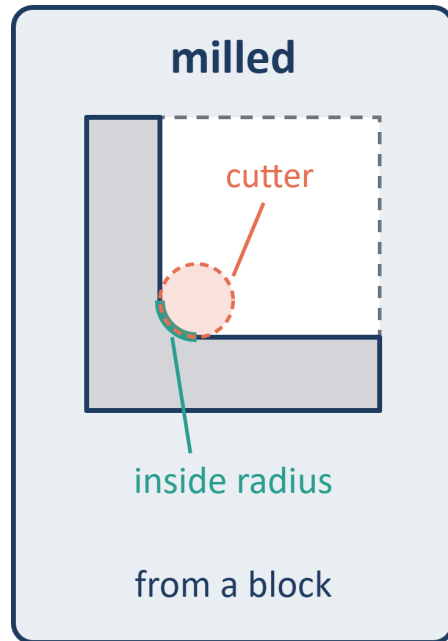


From an idea to a file:
design the way it is
made

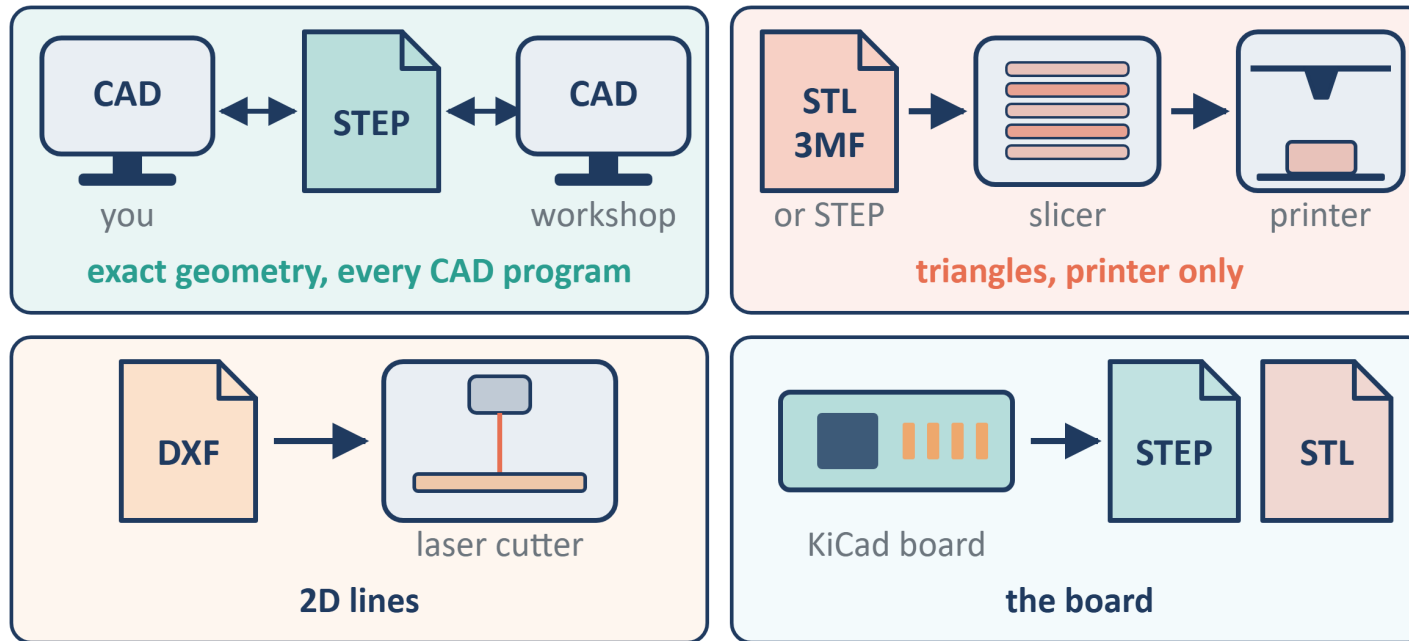


Files · the CAD programs · reading OpenSCAD
· the AI design loop

- The machine decides the shape
- Milled: inside corners keep the cutter's radius
- Bent: bend radii, and a flat pattern to cut first
- Printed: chamfers and flat faces, no overhangs
- Pick the process first, then draw
- The printing rules are the design



- STEP: exact geometry; every CAD program reads it
- STEP is what a machining service wants from you
- STL: triangles only, for the printer
- 3MF: triangles plus colours and print settings
- DXF: 2D lines, for the laser cutter and the water-jet
- KiCad exports the board as STEP (and STL)

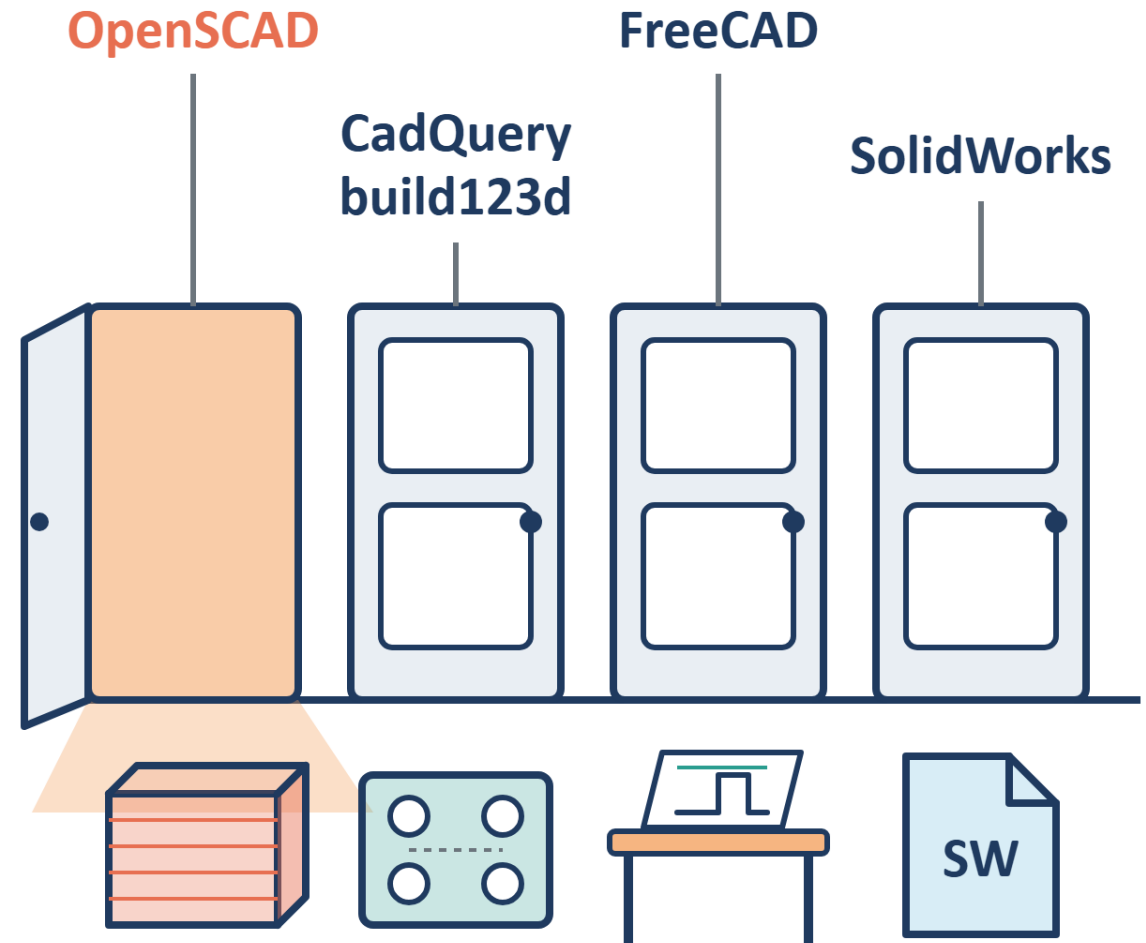


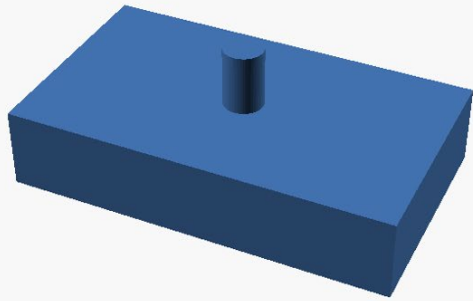
	OpenSCAD	CadQuery	build123d	FreeCAD	SolidWorks
How you drive it	code, own window	Python	Python	GUI + Python	GUI
Cost	free	free	free	free	US\$ 1000s, student licence
Geometry	mesh (facets)	exact	exact	exact	exact
Who uses it	the 3D-printing world	engineers, researchers	newest; AI tooling	the largest free CAD	industry
The AI writes it	best	good	good	mixed	no practical route

- Downloads, manuals and the comparison: the course website, Resources page

	OpenSCAD	CadQuery	build123d	FreeCAD	SolidWorks
Fillets, holes, threads	via the BOSL2 library	yes	yes	yes	yes
Imports the board STEP	no (STL only)	yes	yes	yes	yes
Exports STEP	no	yes	yes	yes	yes
Sketch, sheet metal, drawing	no	sketch only	no	all three	all three
Assemblies	by hand	with a solver	joints	with a solver	with a solver
Where you see the part	own window	OCP viewer, CQ-editor	OCP viewer, CQ-editor	own window	own window

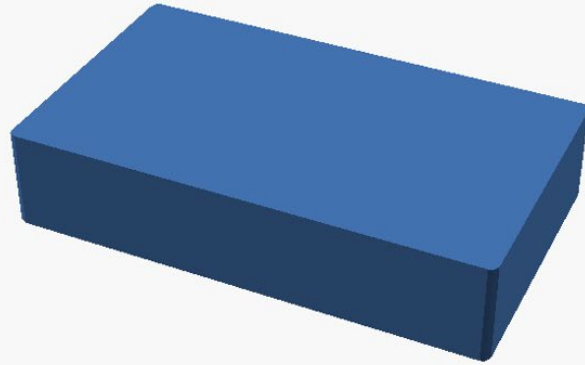
- OpenSCAD, today: a printed box or a laser plate; STL and DXF
- CadQuery, build123d: exact holes and STEP, still code the AI writes
- FreeCAD: drawings for the machinist, sheet metal, FEM; free
- SolidWorks: the industry standard; Onshape: free, in a browser
- The specification and the loop are the same in all five





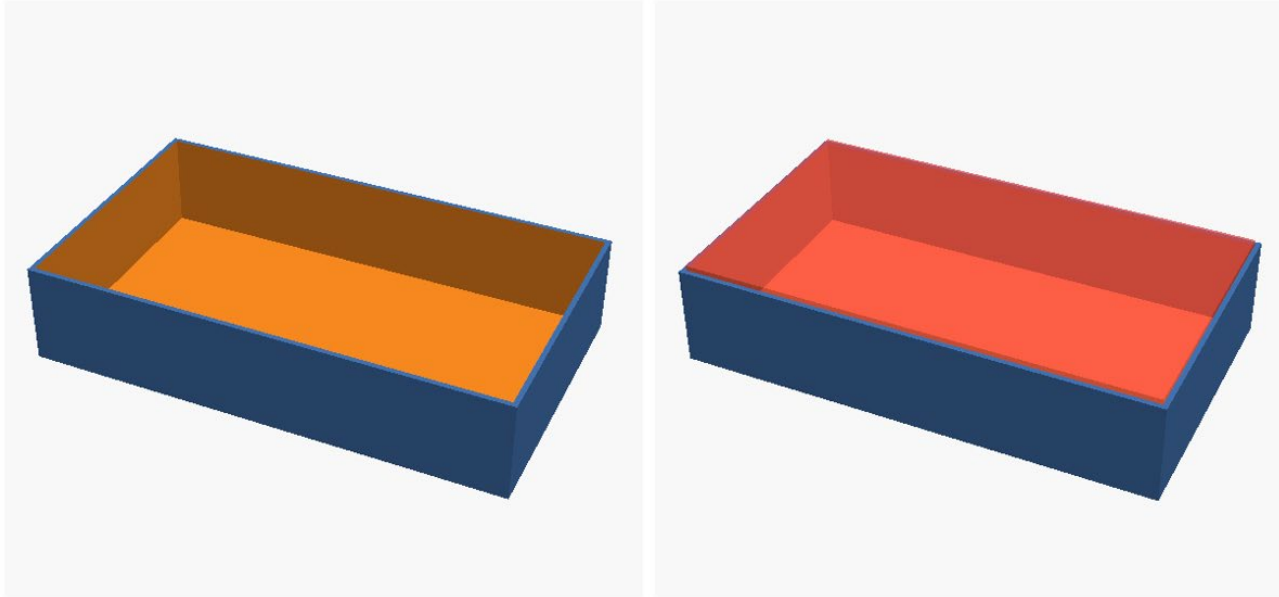
```
// a shape
cube([190, 110, 40]);
// a move in front of a shape
translate([95, 55, 40])
  cylinder(d = 20, h = 30, $fn = 48);
```

- Shapes: cube, cylinder, sphere; square, circle, text
- Moves: translate, rotate, mirror, scale
- Combinations: union, difference, intersection
- A statement: moves in front, the shape, a semicolon
- Save the file and the preview redraws



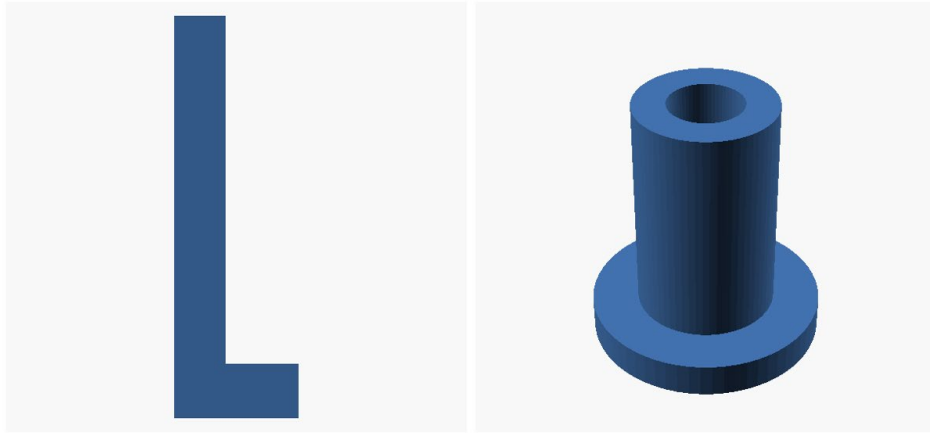
```
linear_extrude(height = 40)  
  offset(r = 3)  
    offset(delta = -3)  
      square([190, 110]);
```

- A 2D shape plus a height is a solid
- square, circle, polygon, text are the sketch
- offset(r) rounds the corners: the mill radius, drawn in
- Read it inside out: the sketch first, then the extrusion
- rotate_extrude() spins a sketch instead: slide D4



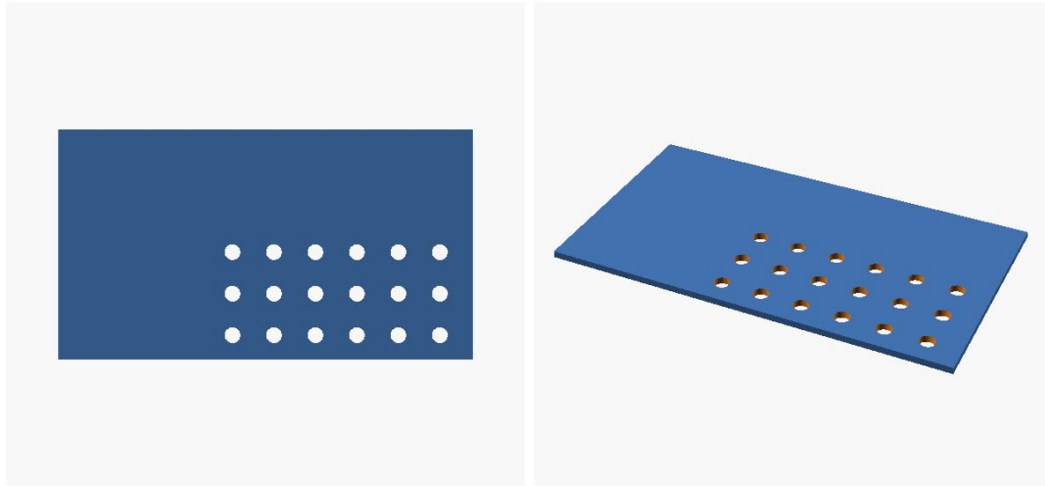
```
difference() {  
  cube([190, 110, 40]);  
  translate([2, 2, 2])  
    cube([186, 106, 40]);  
}
```

- The first shape stays; every shape after it is subtracted
- The pocket is a cube moved in by the wall thickness
- Let a cut overshoot the face, so it cuts cleanly
- The mill removes material; so does difference()
- Put # in front of a cut to see it in red



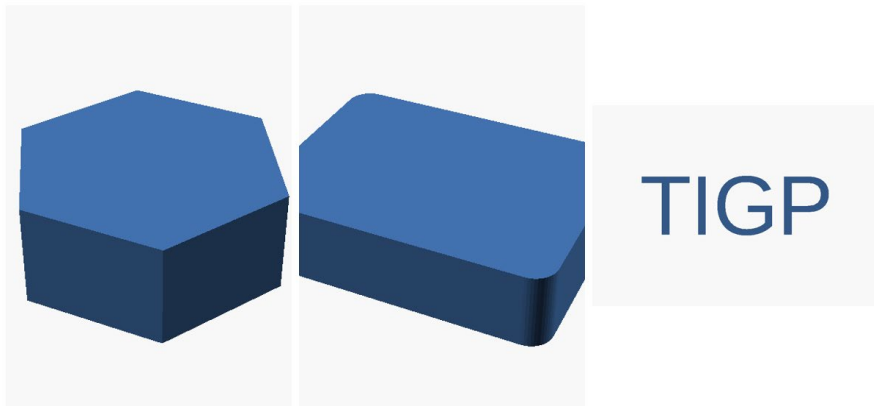
```
rotate_extrude($fn = 64)
  polygon([[1.6, 0], [5, 0], [5, 1.5],
          [3, 1.5], [3, 11], [1.6, 11]]);
```

- A 2D profile right of the axis, spun around Z
- Anything the lathe makes: stand-offs, knobs, bushings
- The inner edge of the profile becomes the bore
- The profile must stay at $x \geq 0$
- A circle moved off the axis and spun makes a ring



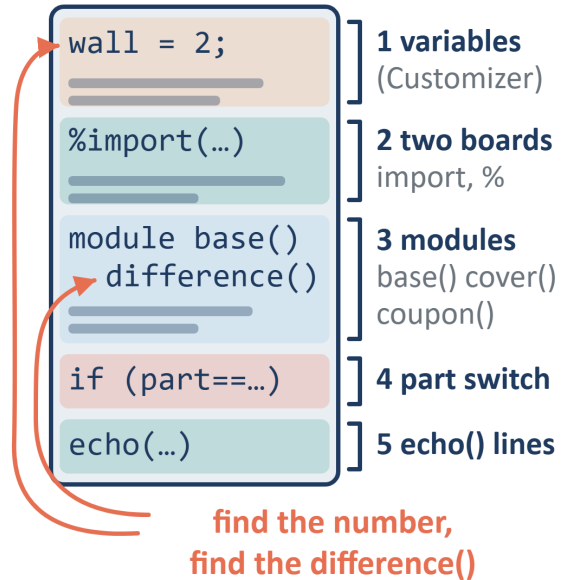
```
difference() {  
  cube([180, 100, 3]);  
  for (x = [75.65:18:165.65],  
      y = [10.67, 28.67, 46.67])  
    translate([x, y, -1]) cylinder(  
      d = 6.5 + 0.3, h = 5, $fn = 48);  
}
```

- A hole is a cylinder you subtract
- for() over the grid writes every SMA hole in one line
- +0.3 mm: the printing rule from earlier, in the code
- A circle is a polygon: \$fn sets its number of sides
- Too few sides and the hole comes out small



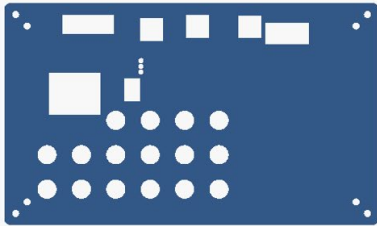
```
cylinder(d = 5.8 / cos(30), h = 2.5,  
         $fn = 6);  
hull() for (x = [3, 37], y = [3, 27])  
  translate([x, y]) cylinder(r = 3, h = 10);  
linear_extrude(0.6) text("TiGP", size = 8);
```

- No threads: a hexagonal nut pocket; the thread is a note
- No 3D fillets: hull() of cylinders, or offset() in the sketch
- No STEP: STL for the printer, DXF from projection()
- Lettering: text() extruded 0.6 mm
- The BOSL2 library adds rounded boxes, screw holes, threads



```
wall=2;           // [1.6:0.2:3]
part="all";       // [all,base,cover,section]
module cover() cube([196, 116, wall]);
%import("class-board.stl");
if (part == "cover") cover();
echo("outer size", 196, 116);
```

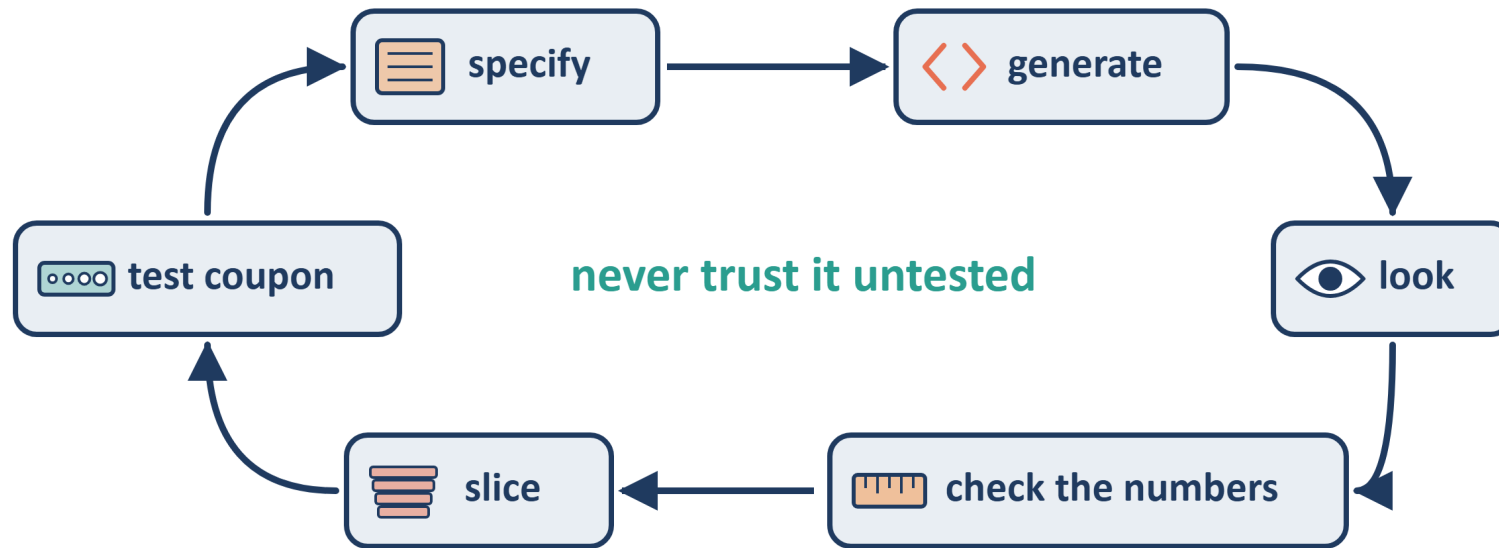
- Variables at the top; a comment there becomes a slider
- A module is a named part
- The part switch chooses what is drawn
- % shows a shape transparent (the boards); # highlights one
- echo() prints the numbers you would otherwise measure
- Reading a file: find the number, find the difference()



```
part = "cover2d"; // [.., cover2d, ..]  
if (part == "cover2d")  
    projection(cut = true)  
        translate([0, 0, -15]) cover();  
// F6, then File > Export > Export as DXF
```

- F5 previews; F6 renders the exact geometry; then File > Export
- One part at a time: set part in the Customizer, export its STL
- A plate for the laser: projection(cut = true) makes it 2D, export DXF
- Can't open import file: the mesh path is wrong, nothing is shown
- Current top level object is empty: the part switch matched nothing
- Ignoring unknown variable: a typo; the shape silently vanishes

- The AI's mistakes look like finished parts
- Check with your eyes: rotate it, cut it open
- Check with numbers: the file prints its own dimensions
- Check with plastic: a test coupon before the real part
- A fresh Claude session reviews the model
- Never trust AI without testing and verifying



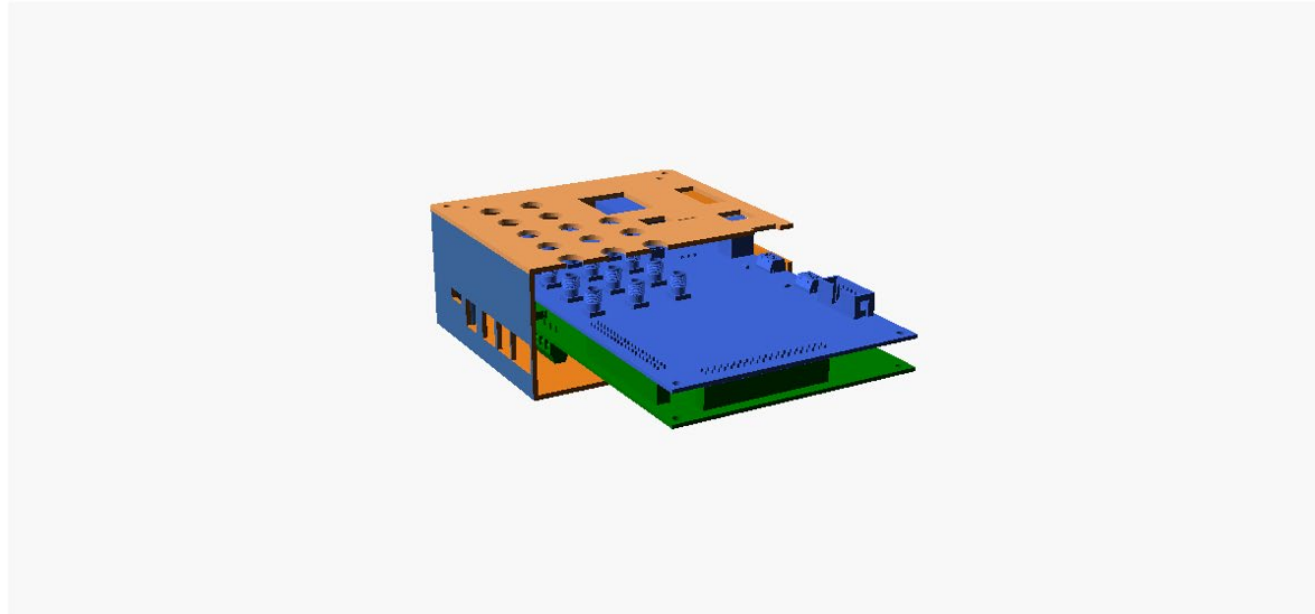
the AI method

vision · specification · review · code · test and verify

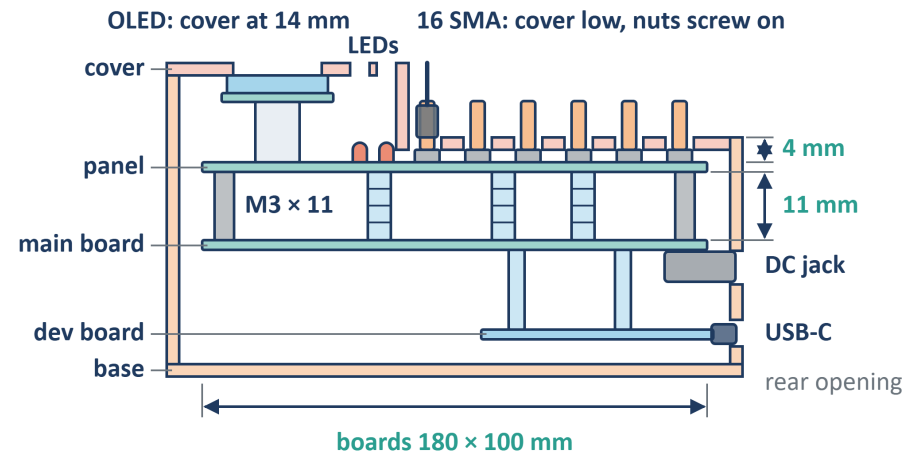
- What it is for, and what it holds
- Sizes, clearances, the faces that lie on the bed
- Fastening, the printing rules, the files you want back
- Last line: “ask me your questions before writing code”

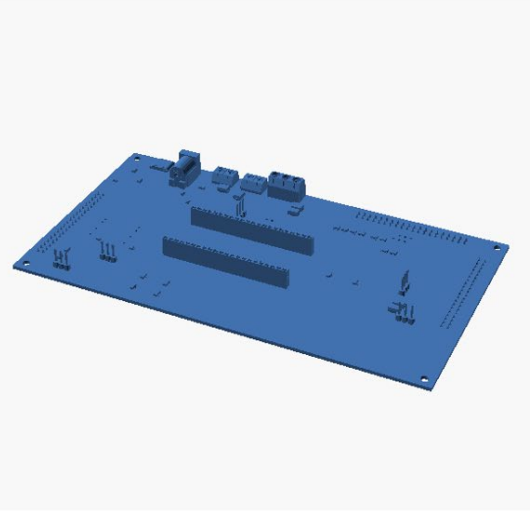
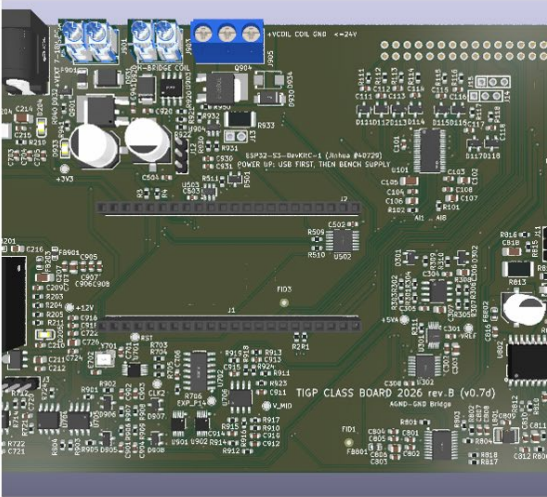
housing specification card	
what it is for	holds two boards, 11 mm apart
the board meshes	two import lines, given
the parts	base · cover · test coupon
the rules	walls 2 mm · 6 mm clearance holes +0.3 · 45° · 256 mm printer
how the file is written	variables · part switch · echo()
make it yours	one change of your own
“ask me your questions before writing any code”	

- Rotate it; the boards inside are transparent (%)
- part = "section" in the Customizer cuts it in half
- The console prints walls, holes, clearance, total size
- # highlights one shape when you distrust a cut
- Bambu Studio: overhangs and supports before printing
- Every finding goes back to Claude as one sentence



- Two boards, one sandwich, 180×100 mm, 11 mm apart
- The panel's outer face is the face of the instrument
- The cover at two heights: low at the SMAs, raised over the OLED
- An SMA plug screws on only if the thread comes through the cover
- Labels: cutouts in the cover, a white sheet behind them
- A rear wall with openings for the cables; a lid that comes off
- Stand-offs and screw lengths set every height
- Every dimension from the board models, none from a ruler

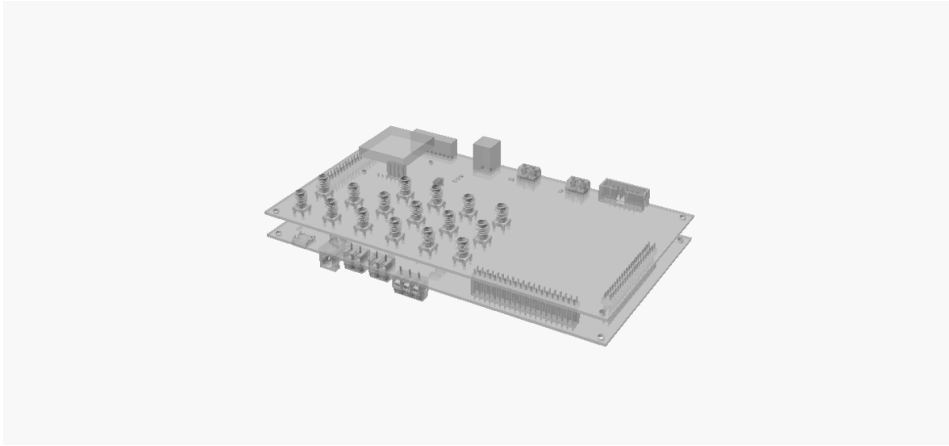




```
kicad-cli pcb export stl  
--subst-models  
--define-var TIGP_BOARD_LIB=<path>  
--component-filter "J*,D*"   
-o class-board.stl  
hardware/class-board.kicad_pcb
```

- KiCad knows every connector's position and body
- Only the parts the housing touches: a light mesh
- One command, shown here over six lines

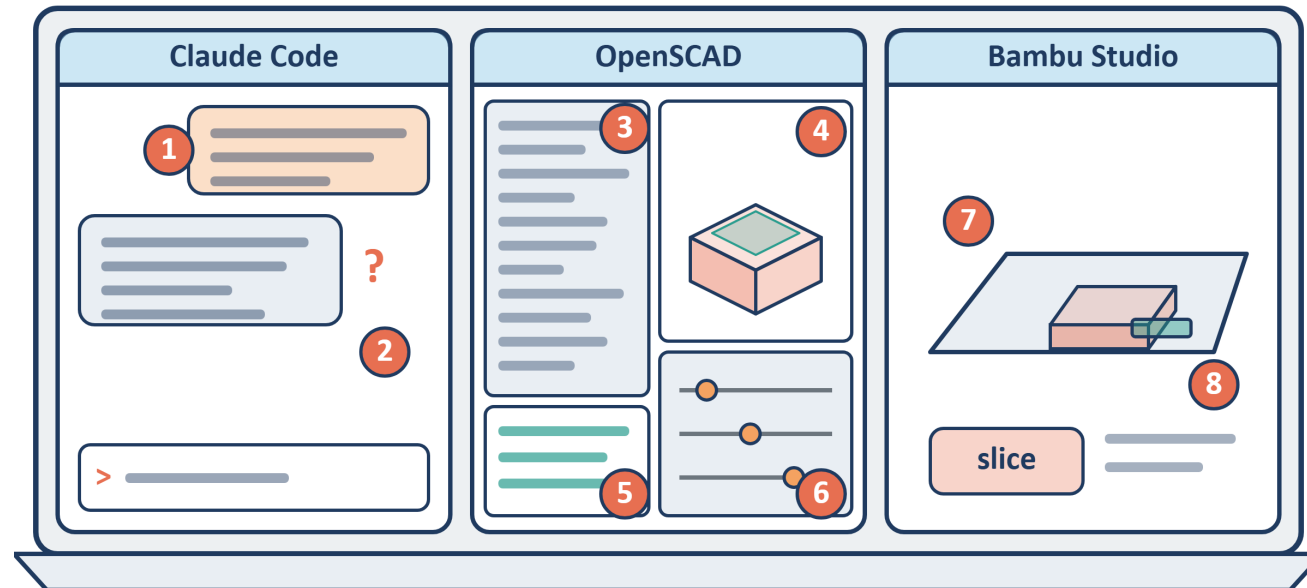
- Made before the workshop, in hardware/release/
- You can re-run it with the tutor



```
// both boards transparent, panel face up
%translate([0, 0, -12.6])
  mirror([0, 1, 0]) mirror([0, 0, 1])
    import("class-board.stl");
%translate([180, 0, -1.6])
  mirror([1, 0, 0]) mirror([0, 1, 0])
    import("front-panel.stl");
```

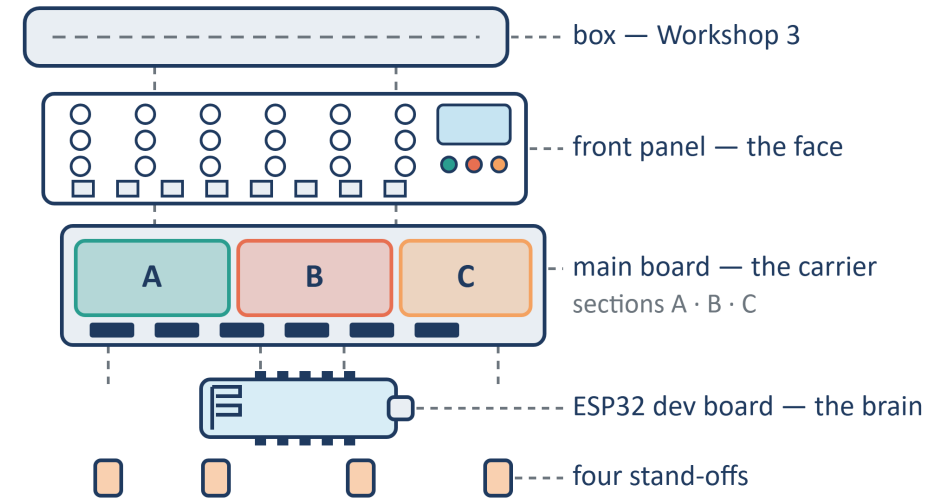
- % shows each board transparent, never printed
- mirror y undoes KiCad's y-down axis; the file does it once
- The main board is turned over: its link sockets face the panel
- The panel mirrored in x is the link mating: $x \rightarrow 180 - x$
- You never type a board coordinate

- 1 Specify: write what the housing must do, end with your questions
- 2 Answer Claude's questions
- 3 Generate: housing.scad appears in OpenSCAD
- 4 Look: rotate, section, find one thing wrong
- 5 Check the numbers in the console against your specification
- 6 Fix it, then make it yours
- 7 Export STL and DXF; slice in Bambu Studio
- 8 Add the test coupon; save and commit your files

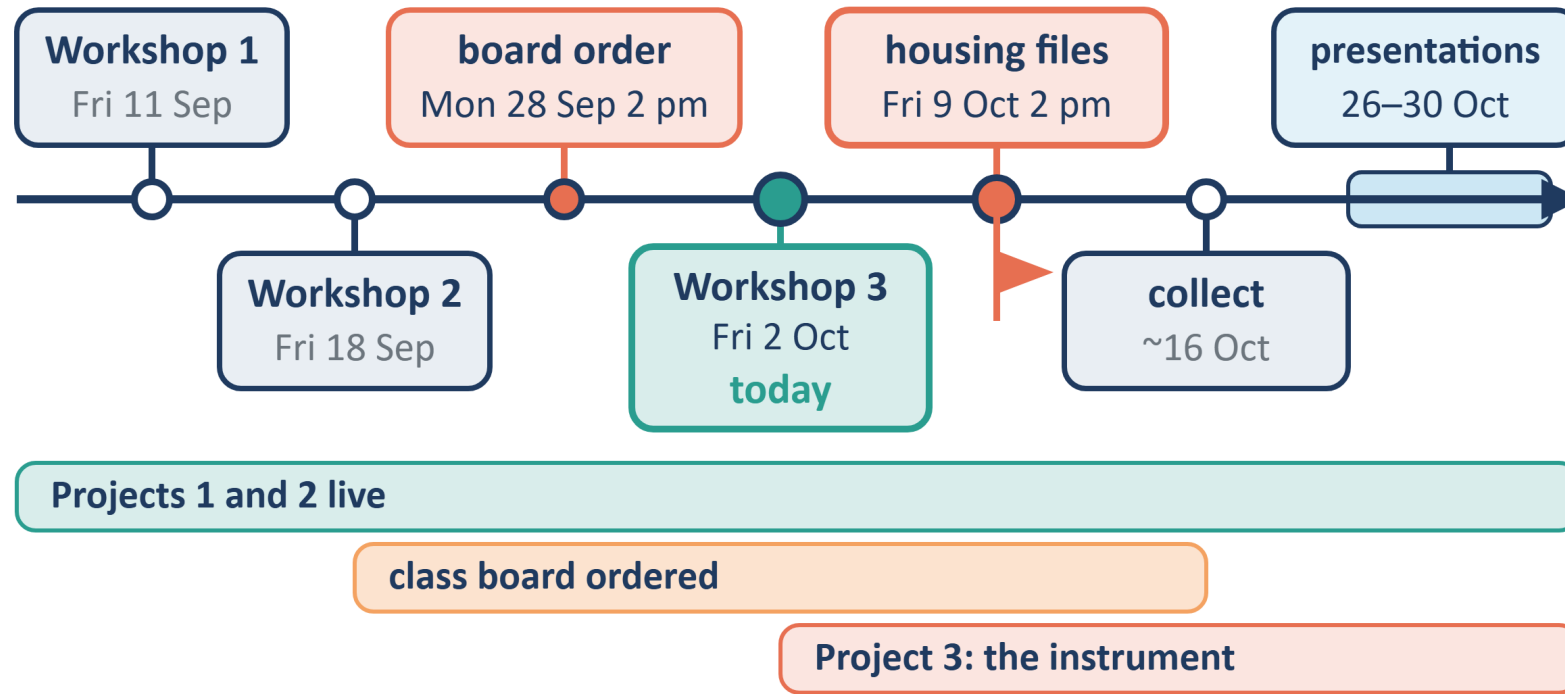


Project 3: build the instrument

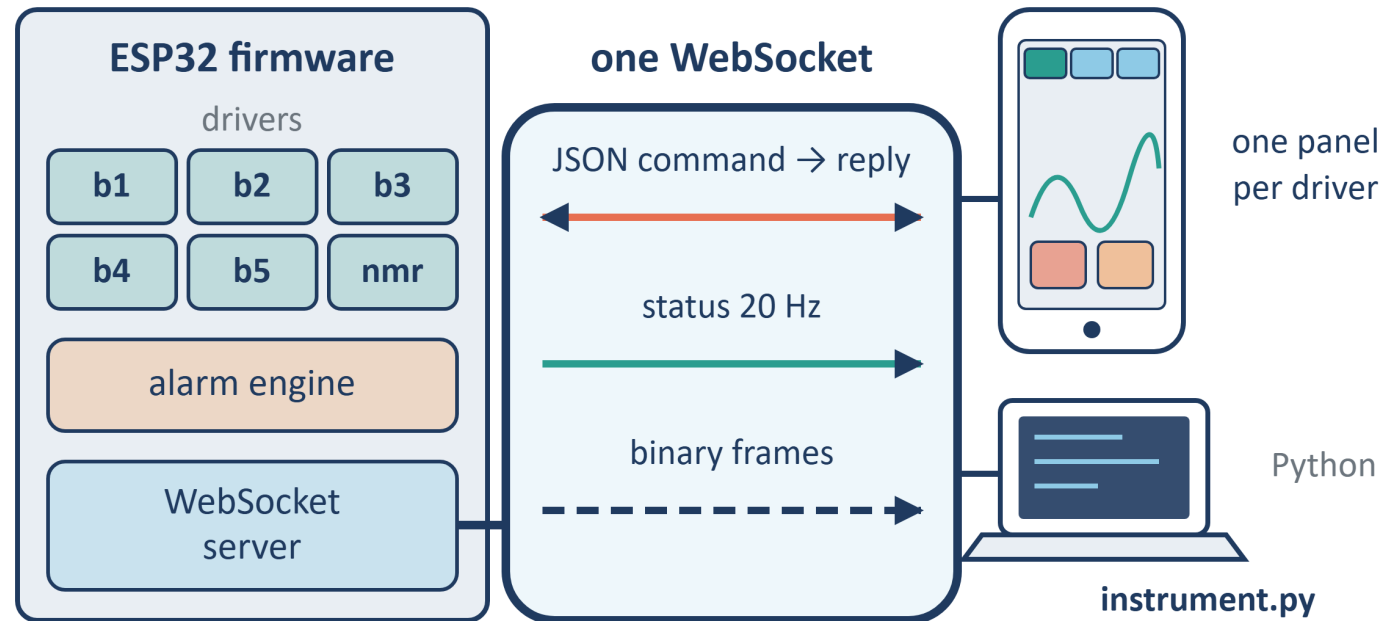
Your firmware and app on the class board, in your housing, on a great website



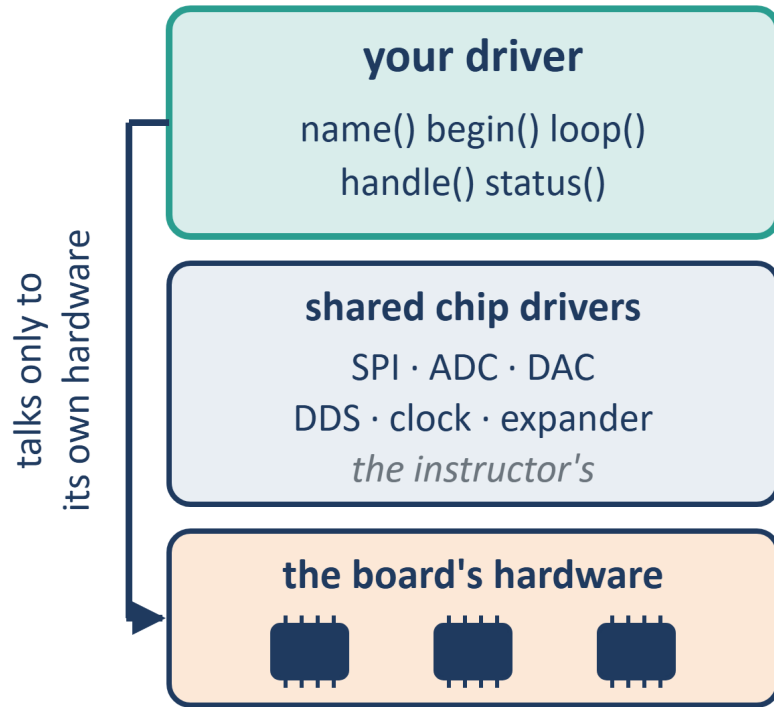
- Projects 1 and 2: live on your websites and on the wall
- The class board, designed together, ordered on 28 Sep: the shared platform
- Today: the housing, and the firmware and app that make the instrument yours
- Still to come: the housing files, your instrument, its website, the talk



- ESP32 firmware: one driver per block, an alarm engine, a server
- JSON command, then exactly one reply
- A status broadcast 20 times a second: the chart, the alarms
- Binary frames for fast data: the stream and the capture
- The same commands from Python: instrument.py on a PC



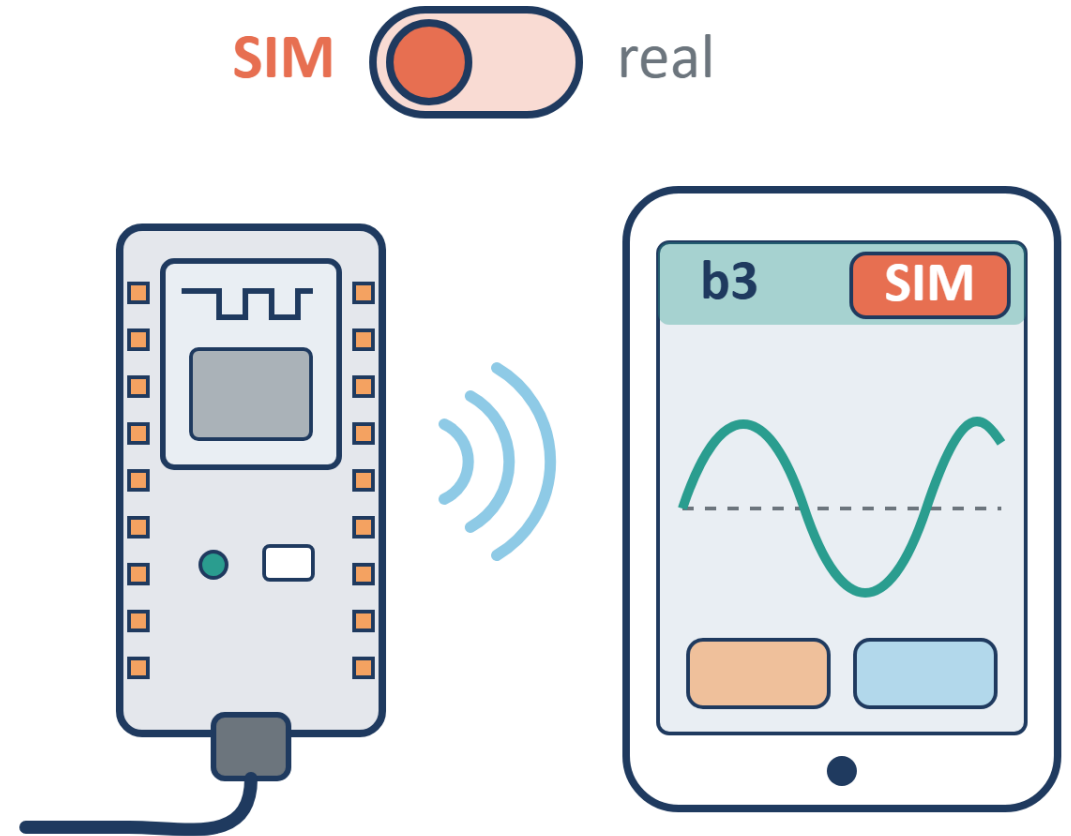
A driver is one class with five methods



```
const char* name() const;
void begin();
void loop();
bool handle(JsonObjectConst cmd,
            JsonObject reply);
void status(JsonObject out);
```

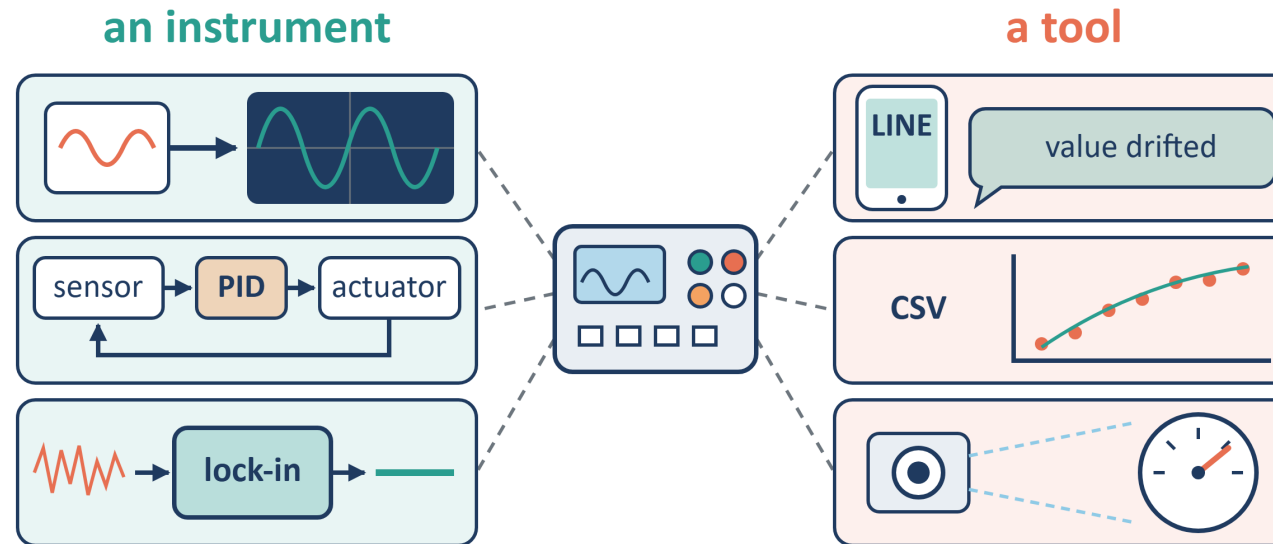
- It talks only to its own hardware
- Everything runs from loop(): no tasks, no locks, no delay()
- Which drivers you write follows your instrument; your section's is the start

- The boards are at the factory; the firmware fakes its hardware
- Your phone or computer talks to the ESP32 exactly as it will later
- Write the SIM branch first: plausible values that move
- Flash esp32s3-sim and watch your status keys on the phone
- The real branch later, from the datasheet and the real board

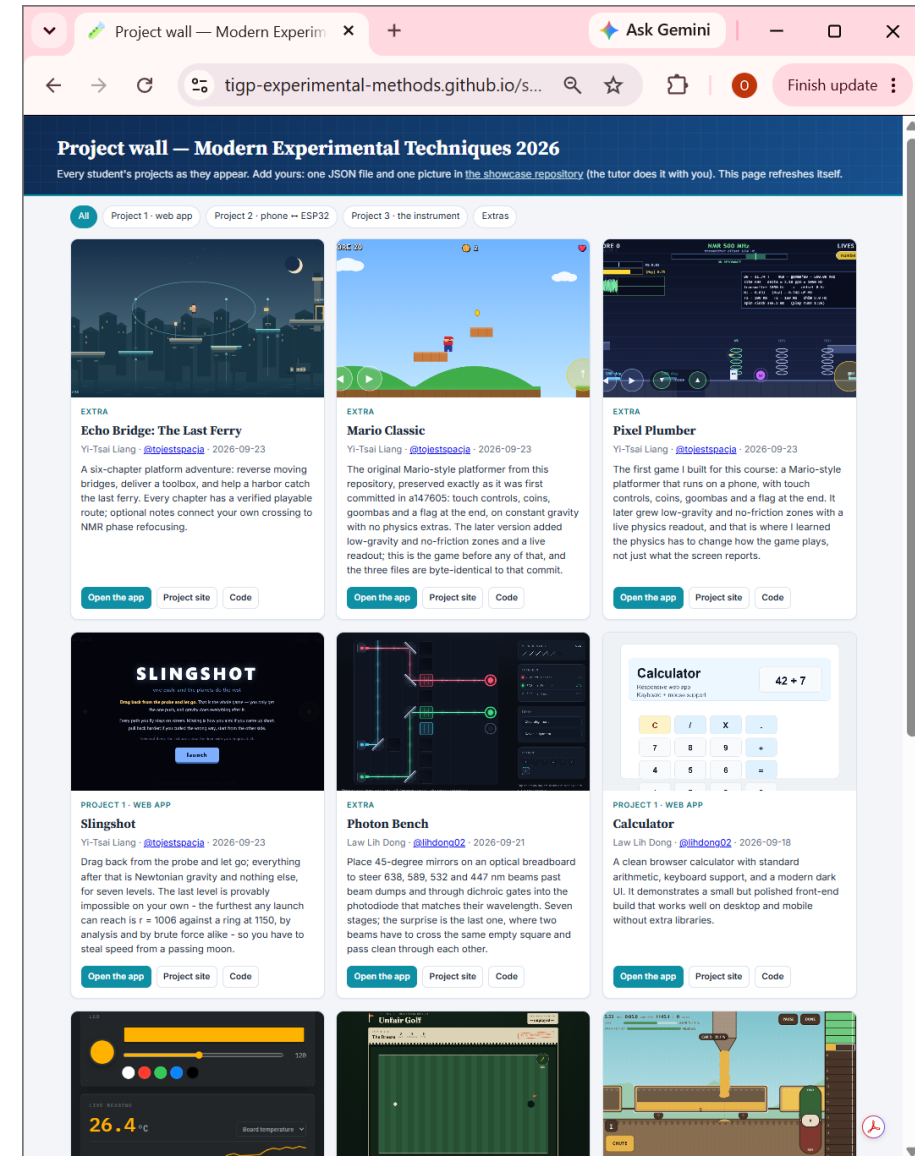


the firmware fakes its hardware

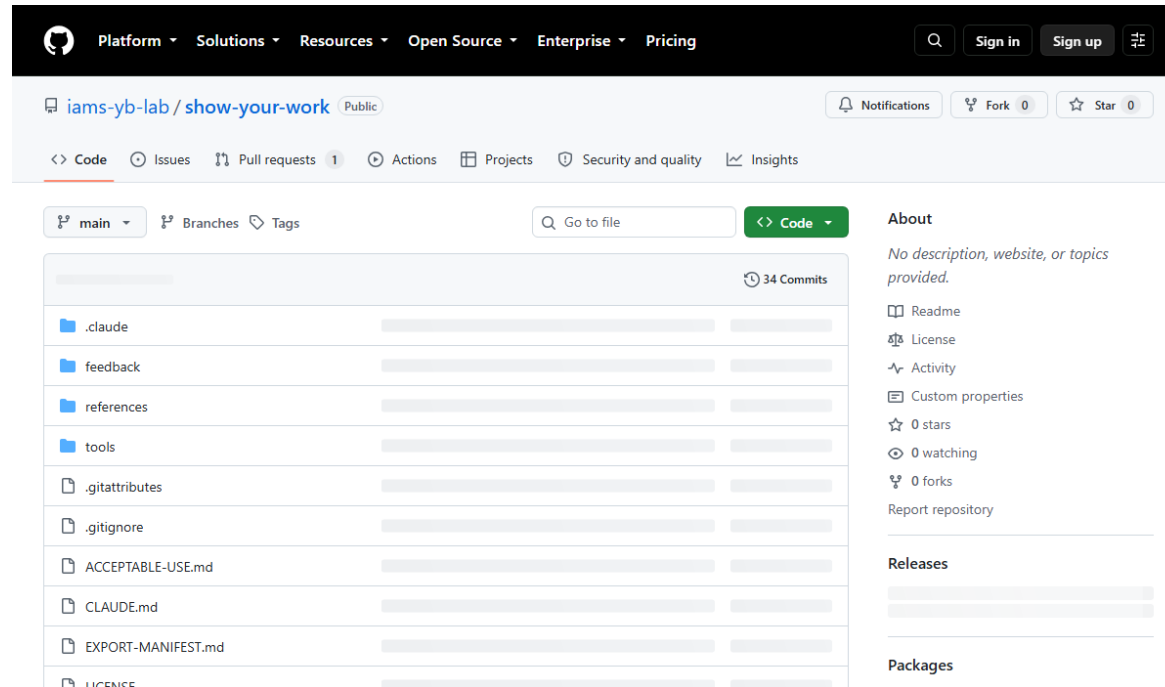
- The board is the shared baseline; what it does is yours
- An instrument: scope and generator, logger, lock-in, PID servo, NMR
- A tool: a LINE or Telegram bot, a Python sweep, a calibration
- A second node: the camera unit watching a gauge
- Can you adapt it to a problem in your research?



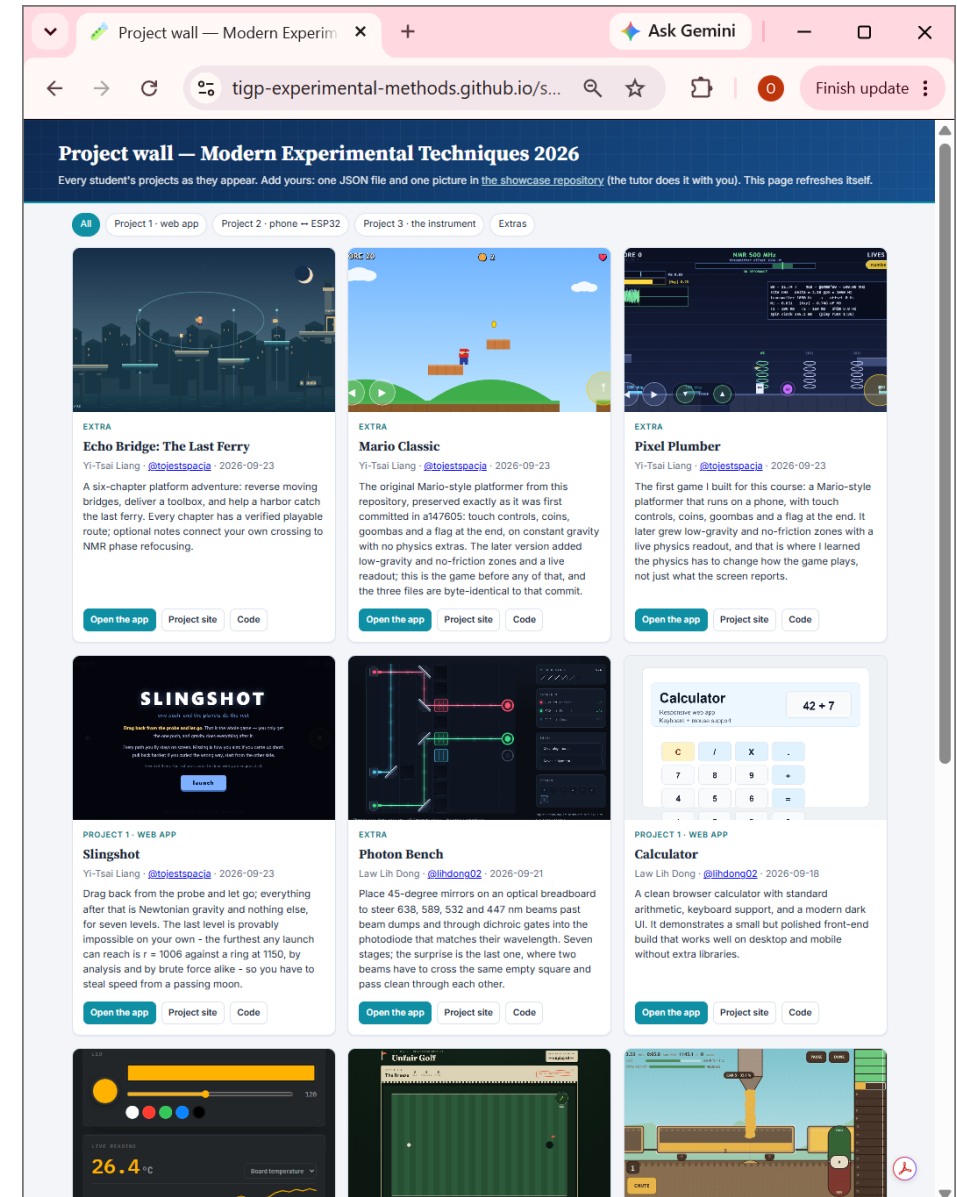
- One repository, one project website, one card on the wall
- The website: what the app does, how to use it, what it can do
- Examples for the user; demonstrations: screenshots, video, animation
- ~16 Oct: boards and housings back; solder the headers; assemble
- Flash without SIM: your app on the real instrument
- 26–30 Oct: present and demonstrate, about 15 minutes each

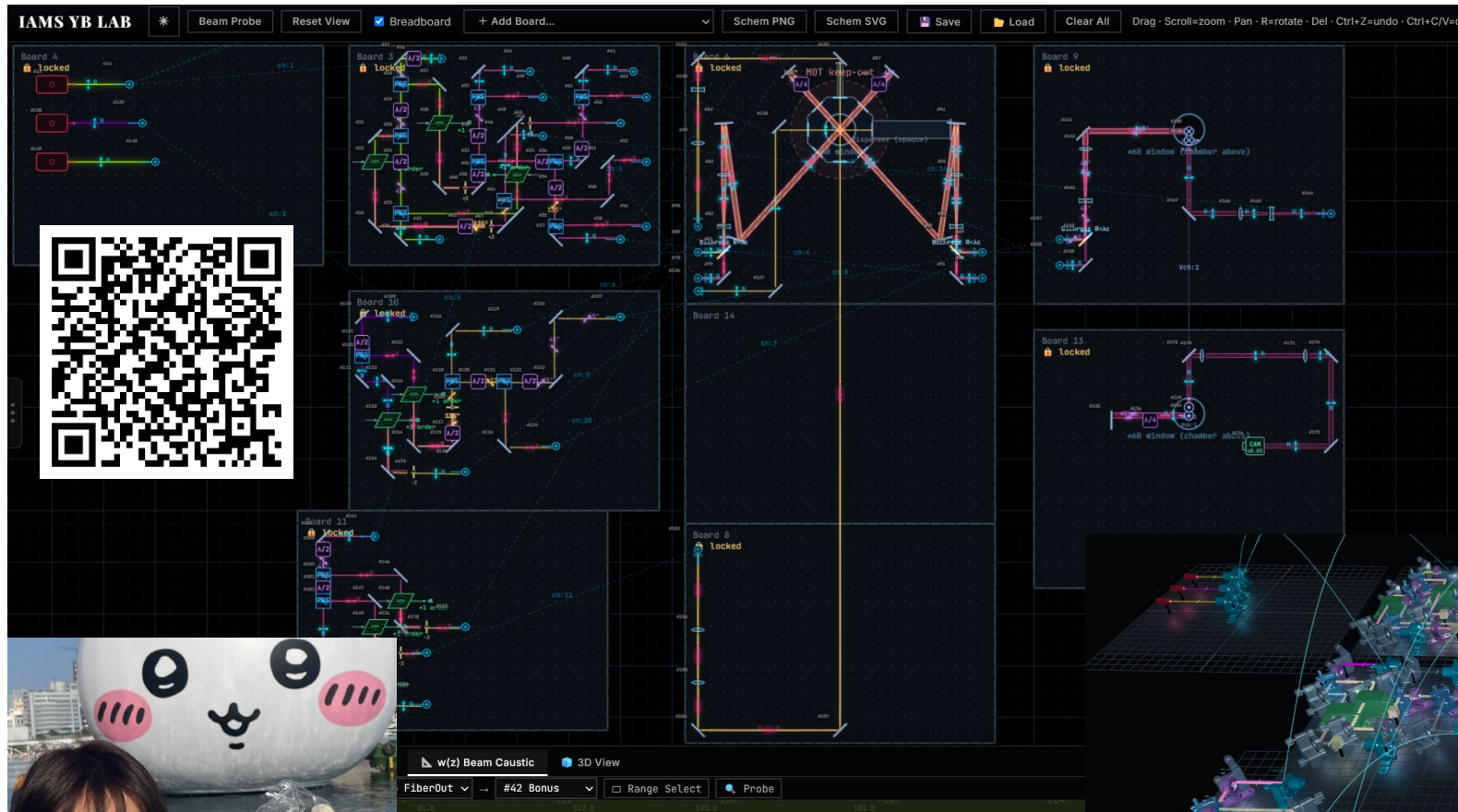


- Optional: a video that presents your app and your instrument
- show-your-work: the AI writes the script and renders the video
- Narration in any language, with subtitles
- The AI can give your project presentation for you, subtitled
- Post it on YouTube and stream it on your project website
- The safety net for the live demonstration



- Housing files by Fri 9 Oct, 2 pm: housing.scad, STL, DXF, a screenshot
- Your firmware and app in sim mode: build the instrument you chose
- ~16 Oct: collect the boards; solder, assemble, flash for real
- 26–30 Oct: present and demonstrate instrument, app, website
- Improve your apps, build and have fun
- Help: the tutor, then
s.p.bennetts@g.iams.sinica.edu.tw



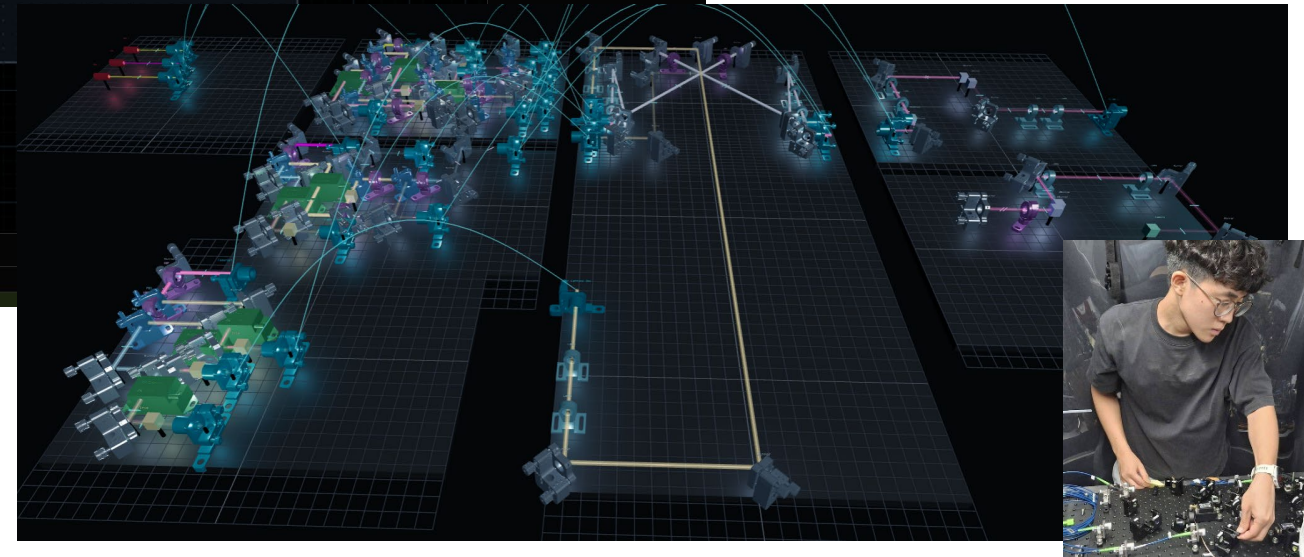


MissAlign

Complex Optical Design
and simulation Tool



Developed (for fun) by
Hao-Rong Yang (Masters student)



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